

LSQ+: Improving low-bit quantization through learnable offsets and better initialization

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Abstract

Unlike ReLU, newer activation functions (like Swish, H-swish, Mish) that are frequently employed in popular efficient architectures can also result in negative activation values, with skewed positive and negative ranges. Typical learnable quantization schemes [5, 7] assume unsigned quantization for activations and quantize all negative activations to zero which leads to significant loss in performance. Naively using signed quantization to accommodate these negative values requires an extra sign bit which is expensive for low-bit (2-, 3-, 4-bit) quantization. To solve this problem, we propose LSQ+, a natural extension of LSQ [7], wherein we introduce a general asymmetric quantization scheme with trainable scale and offset parameters that can learn to accommodate the negative activations. Gradient-based learnable quantization schemes also commonly suffer from high instability or variance in the final training performance, hence requiring a great deal of hyper-parameter tuning to reach a satisfactory performance. LSQ+ alleviates this problem by using an MSE-based initialization scheme for the quantization parameters. We show that this initialization leads to significantly lower variance in final performance across multiple training runs. Overall, LSQ+ shows state-of-the-art results for EfficientNet and MixNet and also significantly outperforms LSQ for low-bit quantization of neural nets with Swish activations (e.g.: 1.8% gain with W4A4 quantization and upto 5.6% gain with W2A2 quantization of EfficientNet-B0 on ImageNet dataset). To the best of our knowledge, ours is the first work to quantize such architectures to extremely low bit-widths.

1. Introduction

With the popularity of deep neural networks across various use-cases, there is now an increasing demand for methods that make deep networks run efficiently on resource-constrained edge-devices. These methods include model pruning, neural architecture search (NAS) and hand-crafted efficient networks made out of novel architectural blocks (e.g. depth-wise separable or group convolutions, squeeze-excite blocks, etc.). Finally we can also perform model quantization, where the weights and activations are quantized to lower bit-widths allowing efficient fixed-point inference and reduced memory bandwidth usage.

Due to the surge in more efficient architectures found with NAS, newer and more general activation functions (like Swish [22], H-swish [11], Leaky-ReLU) are replacing the traditional ReLU. Unlike ReLU, these activation functions also take over values below zero. Current state-of-the-art quantization schemes like PACT [5] and LSQ [7] assume unsigned quantization ranges for activation quantization where all the activation values below zero are discarded by quantizing them to zero. This works well for traditional ReLU-based architectures like ResNet [10], but leads to a significant loss of information when applied to modern architectures like EfficientNet [26] and MixNet [27], which employ Swish activations. For example, LSQ achieves W4A4 quantization of preactivation-ResNet50 with no loss in accuracy but leads to a 4.1% loss in accuracy when quantizing EfficientNet-B0 to W4A4¹. Naively using a signed quantization range to accommodate these negative values also results in a drop in performance.

To alleviate these drops in performance which are commonly observed with very low-bit (2-, 3-, 4-bit) quantization, we propose using a general asymmetric quantization scheme with a learnable offset parameter as well as a learnable scale parameter. We show that the proposed quantiza-

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¹WxAx quantization indicates quantizing the weights and output activations of all layers to x bits